

Ryan Dielhenn

Los Angeles, CA

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Technical Skills

- Languages** Rust, Go, Java, Scala, C, Python, SQL
- Systems** Docker, Kafka, Prometheus, Git, Linux, CI/CD
- Concepts** Distributed Systems, LSM-Tree Storage Engines, Object Storage (S3), API Design, High-Performance Data Processing

Projects

SlateDB & OpenData

OPEN SOURCE CONTRIBUTOR

February 2026 – Present

- Designed and implemented distributed compaction for SlateDB (Rust, LSM-tree on object storage), enabling parallel compaction across multiple workers to address a single-compactor throughput bottleneck; core implementation merged to main.
- Authored the RFC for the design, covering work scheduling, coordinator/worker split, and failure-handling semantics; currently implementing heartbeat-based worker failure detection.
- Contributed to OpenData, a collection of databases sharing a unified object storage foundation via SlateDB; authored and iterated on PRs for enhancements to the TSDB including a regression benchmark, Prometheus metadata endpoint, federate endpoint, and scalar argument support for PromQL.

ZephyrCache

DISTRIBUTED SYSTEMS PROJECT

July 2025 – May 2026

- Built a fault-tolerant distributed cache in Go with consistent hashing and replication across ring successors for data durability.
- Exposed a REST API for simple cache and health check operations, along with metrics collection via Prometheus.
- Reviewed and integrated a contributed SWIM-based UDP gossip protocol for decentralized peer registration, failure detection, and membership management as a lightweight alternative to etcd.

Experience

Confluent

Mountain View, CA

SOFTWARE ENGINEER

January 2021 – July 2022

- Contributed to Apache Kafka's migration from ZooKeeper to KRaft, improving observability and reliability of the distributed consensus layer in Java/Scala.
- Built metrics pipelines in Java/Scala to monitor cluster health, quorum state, and inter-broker communication for KRaft, enabling faster incident detection for customer-facing clusters; updated Confluent Cloud tooling in Go to integrate new metrics.
- Contributed to Confluent Cloud's Cluster Linking integration with KRaft architecture for cross-cluster replication.

Confluent

Mountain View, CA

SOFTWARE ENGINEERING INTERN

May 2020 – August 2020

- Implemented dynamic client reconfiguration in Java/Scala for Apache Kafka, enabling runtime updates to producer/consumer settings (including connection, security, retry, and ack configurations) without service restarts.
- Enhanced Confluent Cloud's rebalance tooling with asynchronous replica movement support.
- Continued contributing to Apache Kafka during Fall 2020 while completing undergraduate degree.

University of San Francisco

San Francisco, CA

UNDERGRADUATE RESEARCHER

January 2020 – May 2020

- Developed a C implementation of Geopresence, a bitmap-based geospatial indexing system using RoaringBitmap compression and HyperLogLog++ for efficient location queries on IoT devices.
- Implemented C-based adaptive grid index achieving 17x speedup over Java; outperformed R-tree indexing by 400x at scale (1M+ points) while R-trees showed advantages on sparse datasets (<7K points).

Education

California State University, Los Angeles

Los Angeles, CA

M.S. IN COMPUTER SCIENCE (IN PROGRESS), GPA: 4.0

Expected December 2026

- Thesis:** Trust and Byzantine Fault Tolerance in Drone Swarms.
- Designing a physics-based trust mechanism that detects Byzantine drones from reported position and movement data, evaluated via event-driven simulation in Java.
- Relevant Coursework: Advanced Computer Networking (BGP, OSPF, TCP/IP), Design and Analysis of Algorithms, Software Engineering, Artificial Intelligence, Machine Learning.

University of San Francisco

San Francisco, CA

B.S. IN COMPUTER SCIENCE, MINOR IN MATHEMATICS, GPA: 3.75

2016 – 2020

- Relevant CS Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms, Big Data, Software Development, Programming Language Paradigms, Senior Capstone
- Relevant Math Coursework: Linear Algebra, Formal Methods (Logic and Proof), Calculus I & II, Modern Algebra